Localizing and Beating the Out-of-Distribution Bottleneck in Vector Search:
A Measured Collection of Routing, Coverage, and Systems Techniques

[TODO: authors]

Abstract

Scalable approximate nearest-neighbor (ANN) search routes a query to a few partition cells, scans quantized codes inside them, and re-ranks a shortlist. On *out-of-distribution* (OOD) workloads — where queries come from a different distribution than the database, as in cross-modal (text→image) retrieval under inner product — we first *localize the bottleneck*: a battery of scan-side and code-side improvements (wider SIMD, higher-resolution and rotated codes, RaBitQ, ADC routing) leave OOD recall/QPS essentially unmoved, while the achievable recall is capped by which cells get *routed to and covered*. We then present a collection of routing- and coverage-directed techniques, each measured in isolation: a one-parameter routing recalibration (γ) grounded in a cap-versus-ball selectivity argument; k-NN-graph coverage augmentation; a joint cross-level partition optimization; multi-assignment; and a corollary that *reverses* standard practice — once graph coverage is present, coarser cells win. Composed and evaluated under a same-hardware, tight-pairwise protocol against ScaNN at its official configuration, the engine attains higher QPS at matched recall@10 at 1M, 10M[pending: , and 100M], and transfers to the streaming track. We report the negative results as first-class evidence for the localization, and treat the measurement methodology — which repeatedly caught mirages during development — as a contribution.

1 Introduction

Approximate nearest-neighbor (ANN) search is the retrieval primitive under retrieval-augmented generation, semantic search, and recommendation. At scale the dominant family is inverted-file with quantization (IVF-PQ): partition the database into C cells, at query time *route* to the $p \ll C$ most promising cells, *scan* compressed codes within them, and exactly *re-rank* a shortlist. A large literature has driven each stage — anisotropic quantization and SIMD fast-scan for the scan, product and residual codes for memory, graph indices as an alternative to IVF — overwhelmingly evaluated where queries and database are drawn from the *same* distribution.

We study the *out-of-distribution* (OOD) regime, where the query distribution differs from the database. This is not a corner case: cross-modal retrieval embeds a text query and searches image vectors under inner product, and the big-ANN "text2image" track (image base, text queries, d=200) is exactly this setting. It is also where the strongest systems separate, and — we argue — where the conventional intuition about what to optimize is wrong.

Our organizing claim is a *bottleneck localization*: on OOD workloads the binding constraint is *routing coverage* — whether the true neighbor's cell is among the p probed and its members reachable — not scan throughput or code fidelity. We support this from both sides. A battery of scan- and code-side improvements (wider SIMD, higher-resolution and rotated codes, RaBitQ, ADC routing) leaves OOD recall and QPS essentially unmoved (§12); meanwhile a small set of routing- and coverage-directed techniques each yields a measured gain (§6–§10). We therefore present the work as a measured collection — every technique reported with the effect it produces in isolation, and the negatives kept as first-class evidence for the localization. Composed and evaluated under a same-hardware, tight-pairwise protocol against ScaNN at its best configuration, the result attains higher QPS at matched recall@10 at 1M and 10M [pending: (and 100M)], and transfers to the streaming track.

Contributions (each measured in isolation).

- **A bottleneck localization** (§5): scan- and code-side levers do not move OOD; routing/coverage does. The negatives (§12) are the evidence.
- γ **routing recalibration** (§6), grounded in a cap-vs-ball selectivity argument (§4). *Effect:* $\approx$ *half the probes at matched OOD recall, zero query-time cost.*
- k**-NN-graph coverage** (§7). *Effect: composes with* γ*; at 1M moves the ScaNN ratio* $1.18 \to 0.98$.
- **The coarse-cell corollary** (§8). *Effect: with graph coverage present, coarser cells win; at 10M* $1.01 \to 0.85$.
- **Joint cross-level partition optimization (tree-EM)** (§9). *Effect:* $+0.6$–1.0 *recall@10 points at 10M, QPS-neutral.*
- **Multi-assignment (SOAR)** (§10). *Effect: raises the routing recall ceiling to baseline parity.*
- **Systems levers for streaming** (§11): batch-parallel insert (4.5×), low-live adaptive p (*worst step* $0.74 \to 0.94$), and a parallelization fix (7.6× *at 8 threads*).
- **A same-hardware evaluation methodology** (§13) and an honest scope of what the ratios do and do not claim.

2 Background: Standard Building Blocks

Our system is assembled from well-established components; we summarize them here so the contributions in §3–§10 can be stated as deltas against known practice.

Inverted file (IVF). Partition the database into C cells (the Voronoi regions of C centroids). At query time, *route* the query to its $p \ll C$ nearest centroids and scan only those cells. End-to-end recall is capped by whether the true neighbor's cell is among the p probed — the routing question that this paper argues is the OOD bottleneck.

Hierarchical routing. Scoring all C centroids per query is $O(Cd)$, which dominates at scale (fine C is needed to keep cells small). A *tree* of centroids — coarse k-means over C_0 centroids, then fine k-means within each — reduces routing to $O(L\,C^{1/L})$ for an L-level tree (FAISS uses an HNSW graph over the centroids for the same purpose). We use hierarchical k-means.

Product quantization (PQ). Split each vector's d dimensions into m subspaces and quantize each subspace to a small codebook (16 codewords for 4-bit PQ). Approximate distances are computed by lookup-and-add over per-subspace tables (ADC). With 4-bit codes the table lookup is a register shuffle (`vpshufb`), enabling the SIMD "fast-scan" kernels (Quick-ADC, ScaNN, FAISS FastScan). *Anisotropic* / score-aware quantization (ScaNN) weights the quantization loss to preserve inner-product ranking rather than reconstruction.

Exact re-ranking. The quantized scan is used only to select a shortlist; exact distances (int8 or float, read for the few shortlisted vectors) re-order it. The shortlist depth t trades recall for cost.

Multi-assignment. Replicating each point into its top-a_0 cells (a space/time tradeoff) raises the chance a query's neighbor lands in a probed cell.

The OOD workload. The big-ANN OOD track (text2image) pairs *image*-embedding base vectors (d=200) with *text*-embedding queries under inner product, with a float ground truth. Base norms are concentrated but not constant ($\approx$ 0.81–0.99); queries are unit-norm. This distribution gap between base and queries is what breaks L_2-trained routing (§6).

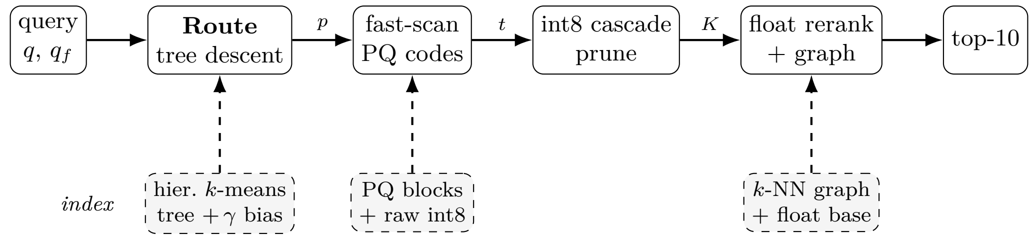

Figure 1: Query pipeline (top) and the offline-built index it consumes (bottom, dashed). Edge labels are the shrinking candidate set: p probed leaves → top-t pool → K cascade survivors → top-10. γ calibration, the k-NN graph union, and the coarse-cell tree geometry are this work's contributions (Table 1).

3 System Overview

Figure 1 shows the pipeline; we describe the build and query paths, then delineate standard vs. contributed components (Table 1).

Build path.

1. **Quantize.** Float base → int8 via a single global scale ($127/\max|x|$); the int8 base is the native scan/route representation.
2. **Train the routing tree.** Hierarchical k-means: C_0 coarse centroids, then fine centroids within each coarse cell to K_f leaves total (2-level; generalizes to L levels).
3. **Joint refinement (tree-EM, §9).** Optionally re-optimize all levels jointly (E: reassign sample points to their nearest leaf via the beam descent; M: recompute every level's centroids).
4. **Multi-assign (SOAR, §10).** Assign each point to a_0 leaves with spill that covers the residual direction of the first.
5. **Encode.** 4-bit anisotropic PQ ($m{=}d/2$ subspaces); pack codes into cell-contiguous 16-row blocks laid out for the `vpshufb` fast-scan. Store the raw int8 vectors in cell order too (cache-warm re-rank gathers).
6. **Calibrate & augment.** Precompute the per-cell γ routing bias (§6); build the base-side k-NN graph sidecar (§7).

Query path.

1. **Route (tree descent).** Score the C_0 coarse centroids (int8, VNNI kernel), keep the top-b_0; expand to their fine children, score those with the γ bias added, and return the top-p leaves.
2. **Scan.** Fast-scan the 4-bit PQ codes of the p probed cells (`vpshufb` ADC with int16 accumulation for ∼12-bit selection resolution; optional AVX-512 64-wide), collecting a bounded top-t pool.
3. **Cascade re-rank.** Deduplicate the pool by original id (needed under multi-assignment); exact-rescore the t survivors in int8 (VNNI dot); prune to the top-K ($K{=}16$).
4. **Float re-rank + graph union.** Compute exact float inner products for the K survivors (reading the original float vectors) unioned with the k-NN graph neighbors of the top-M; return the top-10.

Algorithms 1–3 state the pipeline precisely. All routing scores use the calibrated form $s_\gamma(q,c) = \gamma\|c\|^2 - 2\,q{\cdot}c$ (§6); the constant $\|q\|^2$ is dropped, and smaller s_γ means a better cell. Leaf centroids are $\{c_j\}_{j=1}^{K_f}$ with precomputed bias $b_j = (\gamma-1)\|c_j\|^2$ so that $s_\gamma(q,c_j) = \|c_j\|^2 - 2q{\cdot}c_j + b_j$ is one dot product plus an additive constant.

Algorithm 1: Build

Input: float base $X_f \in \mathbb{R}^{n \times d}$; tree sizes C_0, K_f; multi-assign a_0; calibration γ; EM rounds R; graph degree k

Output: index $(\{Q_0\}, \{c_j\}, b, \text{PQ blocks}, \text{raw}, G)$

$\sigma \leftarrow 127/\max_i \|X_f[i]\|_\infty$; $X \leftarrow \text{int8}(\sigma X_f)$; `// global-scale quantize`

$\{Q_0\} \leftarrow k$-means(X, C_0); $\{c_j\} \leftarrow$ fine k-means within each coarse cell to K_f leaves;

for *R rounds* `// tree-EM (§9); optional` **do**

  E: assign each sample to its nearest leaf via `Route`$(\cdot, 1)$;

  M: recompute $\{Q_0\}, \{c_j\}$ from the assignments;

foreach *point x_i* `// SOAR multi-assign (§10)` **do**

  assign x_i to its top-a_0 leaves; spill covers the residual of the 1st;

per leaf: encode members as 4-bit anisotropic PQ, pack into 16-row `vpshufb` blocks; store raw int8 in leaf order;

$b_j \leftarrow (\gamma - 1)\|c_j\|^2$ for all j ; `// routing bias`

$G \leftarrow$ base-side k-NN graph (self-search; §7);

return index $(\{Q_0\}, \{c_j\}, b, \text{PQ blocks}, \text{raw}, G)$;

Algorithm 2: Route (tree descent) – returns the p best leaves

Procedure `Route`$(q,\ p)$**:**

  $S_0 \leftarrow$ top-b_0 coarse cells minimizing $s_\gamma(q, Q_0)$; `// int8/VNNI kernel`

  $\mathcal{L} \leftarrow$ fine children of S_0;

  score each $c_j \in \mathcal{L}$ by $\|c_j\|^2 - 2q \cdot c_j + b_j$;

  return the p leaves of $\mathcal{L}$ with smallest score;

4 Theoretical Grounding (Scoped)

[TODO: (i) Cap-vs-ball under concentration: additive ball filters are non-selective in high dimension (everything within $R + r$), angular/spherical-cap filters remain selective; k-means on centered data realizes a data-dependent angular partition. (ii) This frames γ: the routing score $\gamma\|c\|^2 - 2q \cdot c$ interpolates γ=1 (additive ball / L_2) to $\gamma \to 0$ (pure inner product / cap), so the theory predicts γ<1 is needed precisely in the concentrated OOD regime – which we confirm. (iii) Two principles we build on: route-precise/rank-cheap, and multi-assignment as a space/time tradeoff. Explicitly state we do NOT claim the recursive-tree polylog result (refuted at $\leq$10M scale).]

5 Localizing the OOD Bottleneck

We separate two ceilings. The *pool-recall ceiling* is the recall obtainable by exactly re-ranking the *entire* probed pool — it depends only on routing (which cells, hence which points, are reachable), not on the scan's approximation. The *scan-limited recall* is what the actual (quantized) scan plus shortlist delivers. If the two coincide, the scan is already extracting everything the routed pool contains and there is no scan headroom; recall is then a pure function of routing coverage.

On OOD text2image this is what we observe: at a fixed probe budget the scan-limited recall sits at the pool-recall ceiling, and that ceiling — not the scan — is what moves when we change routing. Two consequences follow, and the rest of the paper is their systematic exploration. First, techniques that improve *coverage* (route calibration, graph augmentation, joint partition optimization, multi-assignment) should raise recall; §6–§10 confirm each does, with a measured effect. Second, techniques that only improve the *scan or the code* (wider SIMD, higher-resolution or rotated codes, alternative quantizers, ADC routing) should not help, because there is no scan headroom to recover; §12 confirms each does not. The negatives are thus not incidental — they are the control that localizes the bottleneck to routing coverage.

Algorithm 3: Query

Input: int8 query q, float query q_f; probes p; pool depth t; prune K; graph fan-in M

$\mathcal{C} \leftarrow$ `Route`(q, p);

$\mathcal{P} \leftarrow \emptyset$; `// bounded top-`t` pool`

foreach *cell* $\in \mathcal{C}$ **do**

  fast-scan its 4-bit PQ codes (`vpshufb` ADC, int16 accum) $\rightarrow$ approx. scores;

  keep the t best in $\mathcal{P}$ (threshold-bounded collect);

$\mathcal{P} \leftarrow$ dedup $\mathcal{P}$ by original id ; `// needed under `$a_0 > 1$

$\mathcal{K} \leftarrow$ top-K of $\mathcal{P}$ by exact int8 IP (VNNI dot) ; `// cascade prune`

$\mathcal{R} \leftarrow \mathcal{K} \cup \bigcup_{r \in \text{top-}M(\mathcal{K})} G[r]$; `// `k`-NN-graph coverage union`

return top-10 of $\mathcal{R}$ by exact float IP $\langle q_f, x_r \rangle$;

Table 1: Components of the pipeline: standard building blocks vs. this work.

Component	Basis	Our delta
int8 IVF partition	standard	—
Hierarchical k-means tree	standard (FAISS-style)	fast int8/VNNI descent kernel
4-bit anisotropic PQ	ScaNN / FastScan family	int16-resolution + 32/64-wide fast-scan
Multi-assignment (SOAR)	ScaNN-derived	dedup-before-cap correctness fix
int8→float re-rank cascade	folklore	the specific K=16 composition
γ routing calibration	**ours**	the OOD miscalibration fix
k-NN-graph coverage union	**ours**	composition with γ + float re-rank
Coarse-cell geometry knee	**ours**	reverses "finer is better"
Tree-EM joint optimization	**ours**	cross-level partition objective
Batch-insert / adaptive-p (streaming)	**ours**	§11

6 γ: One-Parameter Routing Recalibration

An IVF router ranks cells by a score over the query q and centroid c. For L_2 k-means the score is $\|q - c\|^2 = \|q\|^2 + \|c\|^2 - 2\,q\cdot c$; dropping the query-constant term leaves $\|c\|^2 - 2\,q\cdot c$. We write the family

$$s_\gamma(q, c) \;=\; \gamma\|c\|^2 - 2\,q\cdot c, \tag{1}$$

where γ=1 recovers L_2 routing and $\gamma\to 0$ recovers pure inner-product (angular) routing. Under distribution shift these disagree in a specific way: the $\|c\|^2$ penalty demotes high-norm centroids, but for OOD queries the inner-product-relevant neighbors concentrate precisely near those high-norm cells, so γ=1 systematically misroutes them. A single global $\gamma<1$ rebalances the two terms. It is grounded in the cap-vs-ball argument of §4: γ slides routing along the additive-ball (γ=1, non-selective under concentration) to spherical-cap ($\gamma\to 0$, selective) axis.

Crucially the fix is *free at query time*: s_γ differs from the L_2 score only by the per-cell constant $(\gamma-1)\|c\|^2$, which we fold into a precomputed integer bias added during cell scoring (no extra query work, no index-format change). We fit γ once per dataset; $\gamma\approx 0.5$ is optimal on text2image at both 1M and 10M, and the recall is flat across $\gamma \in \{0.4, \ldots, 0.6\}$ near the optimum.

Effect. Reaches matched OOD recall at ≈half the probes (p: 54→29 at 1M; ≈2.5× probe cut at 10M). Zero query-time cost.

7 k-NN-Graph Coverage Augmentation

[TODO: Union the graph neighbors of the top-M shortlist members into the exact re-rank set (base-side IP-kNN sidecar; built by ScaNN self-search or engine self-query, ∼92% edge agreement). Recovers deep neighbors a shallow probe missed. Composes additively with γ (partial overlap, compound ~ 0.97).]

Table 2: Tree geometry and per-query routing cost (Eq. 2) by scale. Beams are aggressive ($b_0 \approx 12$–25% of coarse cells) because OOD queries route poorly. [pending: 100M validated via a scaling-law sweep (§8.1).]

scale	L	K_f	cell counts	beams	route evals	pts/leaf
1M	2	16,384	C_0=768	b_0=96	$768 + 96{\cdot}21 = 2{,}816$	61
10M	2	65,536	C_0=4096	b_0=128	$4096 + 128{\cdot}16 = 6{,}144$	152
100M	3	524,288	C_0=2048, C_1=32768	b_0=512, b_1=256	$2048 + 512{\cdot}16 + 256{\cdot}16 = 14{,}336$	190

Effect. At 1M, composed with γ, the ScaNN ratio drops $1.18 \to 0.98$ (the first sub-1 result); at 10M it lets t_{surv} halve at matched recall.

8 Routing Cost and the Coarse-Cell Corollary

8.1 A routing cost model

Consider an L-level tree over K_f leaves with cell counts $C_0 < C_1 < \cdots < C_{L-1} < C_L = K_f$ and per-level beam widths $b_0, \ldots, b_{L-2}$ (the number of cells expanded when descending from level i to level i+1). Routing a query costs, in centroid distance evaluations,

$$\text{route}(q) \;=\; C_0 \;+\; \sum_{i=1}^{L-1} b_{i-1}\,\frac{C_i}{C_{i-1}}, \tag{2}$$

since the coarse level scores all C_0 centroids and level i expands the b_{i-1} best parents from level $i-1$, each contributing C_i/C_{i-1} children. The scan then reads $\text{scan}(q) = p \cdot (n/K_f) \cdot a_0$ candidates (p probed leaves $\times$ points/leaf $\times$ multi-assignment). For a 2-level tree (2) is $C_0 + b_0 K_f / C_0$, minimized at $C_0 = \sqrt{b_0 K_f}$ to give $2\sqrt{b_0 K_f} = O(\sqrt{K_f})$; with L levels and balanced branching this falls to $O(L\,K_f^{1/L})$. The two costs pull oppositely in K_f: finer leaves shrink the scan (n/K_f) but grow the routing. The beams b_i are the "routing aggression" knob — wider beams retain the true cell with higher probability at linear cost — and OOD queries, which route poorly, need wider beams (§14).

Table 2 gives geometry and routing cost by scale. Two levels suffice through 10M; at 100M a 2-level tree at its optimal $C_0 = \sqrt{b_0 K_f} \approx 16{,}400$ would cost $\approx$33,000 evals, whereas the 3-level tree roughly halves this (14,336) — so the third level is introduced only at 100M, where routing is a large enough share of the query to matter.

8.2 Scaling laws for geometry

To set geometry at scale without per-n retuning, we sweep K_f at $n \in \{10^5, 3{\cdot}10^5, 10^6, 3{\cdot}10^6, 10^7\}$ (2-level, graph off), at each n finding the minimum p that reaches recall@10 ≥ 0.90 (deterministic, hence load-independent) and the resulting cost route + scan from Eq. (2). The cost-minimizing geometry follows clean power laws (log-log fit):

$$K_f^\star \sim n^{0.66}, \qquad \tfrac{n}{K_f^\star}\,(\text{pts/leaf}) \sim n^{0.34}, \qquad p^\star \sim n^{0.18}, \qquad C_0^\star \sim n^{0.33}.$$

The central law is pts/leaf $\sim n^{1/3}$: optimal leaves *grow* with n, because routing $O(\sqrt{K_f})$ rises while the scan $O(p\,n/K_f)$ is held down by the growing K_f — the coarse-cell corollary expressed as a scaling law. Probes grow only as $n^{1/5}$. Extrapolating, n=10^9 calls for $K_f \approx 10^6$ ($\sim$1000 pts/leaf). These calibrate the 2-level baseline; graph coverage (§7) shifts the optimum coarser and a third routing level (cheaper, $O(K_f^{1/3})$) shifts it finer, so the 100M geometry is validated by direct A/B (§14) rather than read off the 2-level law.

8.3 The corollary

[TODO: Sweep cell granularity at 10M with the graph on: fine (Kf=262144), medium (65536), coarse (32768); medium/coarse WIN. Bracket the knee with four builds.] The cost model (2) explains *why*. Without graph coverage, recall demands fine K_f (small leaves make the true neighbor's leaf likelier to be among the p probed), paying $O(\sqrt{K_f})$ routing. The k-NN-graph union (§7) supplies that coverage *independently of* K_f, so the tree can use *coarser* leaves — cheaper routing — at fixed recall. The optimum is a knee: too fine wastes routing, too coarse inflates the scan $(p \cdot n/K_f)$. Empirically the knee sits near 150 points/leaf at 10M, and both finer and coarser builds lose to it.

Effect. At 10M, moving from the fine-cell to the 152-point-cell geometry takes the ScaNN ratio 1.01 → 0.85; routing evals/query drop ∼12,000→6,144.

9 Joint Cross-Level Partition Optimization (Tree-EM)

[TODO: EM over the hierarchical partition: E-step reassigns to nearest leaf via the beam descent (objective = query-time search), M-step recomputes every level jointly. Cheap-beam E-step. Additive with γ/graph.]

Effect. +0.6–1.0 recall@10 points at 10M at matched probe, QPS-neutral (build-time only).

10 Multi-Assignment (SOAR)

[TODO: Replicate each point into a_0 cells; SOAR spill covers the residual direction of the first cell. A dedup fix (rerank by orig id before the survivor cap) was required to realize it. Raises the routing coverage ceiling.]

Effect. a_0=2 reaches OOD coverage parity with the baseline partition; higher a_0 extends the high-recall frontier at a storage/scan cost (quantified in table).

11 Systems Techniques (Streaming)

[TODO: msturing-30M-clustered final_runbook; recall@10 under 1h + 8GB. Three levers: (i) batch-parallel insert (assignment parallelized over the batch, order-preserving, bit-identical); (ii) low-live adaptive p (grow probes so expected candidates ≥ threshold when the live set is small); (iii) a parallelization fix (the batched re-rank driver ran chunks serially -> RAYON never engaged). Plus a compliance re-architecture: first-insert-batch cold start + periodic live-set retraining.]

Effect. Batch insert: 30M inserts 1818s → 406s (4.5×), recall-identical. Adaptive p: worst step 0.737 → 0.944. Parallelization fix: 7.6× at 8 threads. Compliant recall@10 = [pending: final NQ=10000], peak <8GB.

12 Negative Results (Evidence for the Localization)

[TODO: Report each with its measured null/loss:]

- **Query-aware routing** (train cells on the query distribution): *worse* than base-trained (0.76 vs 0.89 @ matched p) — undertrained + the cells where queries are dense don't distribute base points well.
- **ADC routing** (4-bit-scan the centroids): recall-neutral but *no speed win* — the exact VNNI router is already at the compute floor.
- **Anisotropic/OPQ-rotated coarse codes for rank preservation**: do *not* make coarse codes rank-preserving; reorder depth is set by code bit-rate, and rotation/η move it < 0.01. Coverage-via-coarsening collapses (a scann-coverage-matched dpb=50 config yields recall 0.09).
- **RaBitQ**: no ranking edge over PQ at matched bytes.
- **Wider SIMD** (AVX-512 vpermw / 64-wide): scan is memory-bound, not shuffle-bound; ≤+7%, sometimes negative (downclock/cross-lane).
- **Finer cells at matched probe**: lose recall (half the probe fraction).

Table 3: OOD text2image, recall@10-matched QPS ratio (ScaNN/ours; < 1 = our engine faster). Same-hardware, tight-pairwise, opponent at official config. [pending: 100M row].

Scale	1-thread	8-thread	recall@10 gate
1M	0.755	0.756	0.905
10M	0.827	0.833	0.901
100M	[pending:]	[pending:]	[pending:]

- **PCA/LeanVec dim reduction on OOD**: near-isotropic embeddings; dead.

These cluster on the scan/code side and all fail; the §5–9 levers cluster on routing/coverage and all succeed — the localization made concrete.

13 Evaluation Methodology

We report *same-hardware ratios*, never cross-machine QPS. All comparisons run on one physical machine; each competitor is measured in the same session, seconds apart, at its own best operating point.

Protocol. For a target recall (recall@10 ≥ 0.90), we (i) fix each system's operating point by an independent sweep, requiring both to clear the recall gate against the *same* exact float ground truth; (ii) alternate the two systems round to round (order flips) so neither owns a load window; (iii) take best-of-k within a round and report the *median of per-round ratios*; (iv) recompute recall from raw ids, never trusting a cached number. On a shared machine, load drift between two builds minutes apart exceeds the effects being measured, so only interleaved same-window ratios and load-independent recall are trusted.

Warm-resident semantics; pristine confirmation. Both systems measured warm and resident (leaderboard serving); an early cold-spawn asymmetry is corrected by scoring each engine's warm block. Headline numbers pause all other first-party jobs (SIGSTOP) and confirm across $\geq$20 rounds.

The baseline at its best. At 10M, where an *official* big-ANN recipe exists, ScaNN uses it verbatim (40k leaves, anisotropic AH/LUT16, residual quant, bfloat16 re-order, top-level partitioner) with its own leaves-to-search / re-order sweep including corner configs. At scales with no official recipe (1M, 100M) we instead give ScaNN its *speed-optimal* configuration, found by sweeping the leaf count: e.g. at 1M, ScaNN peaks at 1200 leaves (6.2k QPS@0.90), with both finer and coarser partitions slower — coarser because routing over more leaves outweighs the finer scan, mirroring our own coarse-cell result (§8). This matters: a convention-chosen leaf count can silently under- or over-partition the baseline. A win depending on a handicapped baseline is quarantined; we describe one (an under-leaved cached 10M index) and its correction.

What we do not claim. Same-hardware ratios on one (shared) machine, not a standardized-VM submission; ScaNN is the *measured* opponent — gaps to binary-only leaderboard systems are *inference* from the public leaderboard.

14 Results

[TODO: Add per-lever isolation deltas; 8-thread scaling; streaming table (recall, peak mem, wall vs 1h budget). Note the ablation ordering is the actual development arc.]

15 Related Work

Table 4: Ablation ladder at 10M (ScaNN/ours ratio; each row adds one lever).

Configuration	ratio
γ only	1.24
+ graph coverage	1.01
+ tree-EM	0.92
+ coarse-cell geometry (knee)	0.85
+ pristine confirmation (20 rounds)	0.83

[TODO: ScaNN (anisotropic VQ), DiskANN/HNSW (graphs), FAISS IVF-PQ, big-ANN challenge, RaBitQ. Position OOD-specific routing as the gap; distinguish γ from residual quantization (orthogonal) and from query-aware training (shown to fail).]

16 Limitations

[TODO: Single (shared) machine; γ fit per dataset; the coarse-cell corollary is demonstrated on IP/OOD data and may not transfer to in-distribution L_2; the gap to binary-only leaderboard systems is inference; filter/sparse tracks untouched; standardized-VM submission is future work and the gating experiment for a leaderboard claim.]